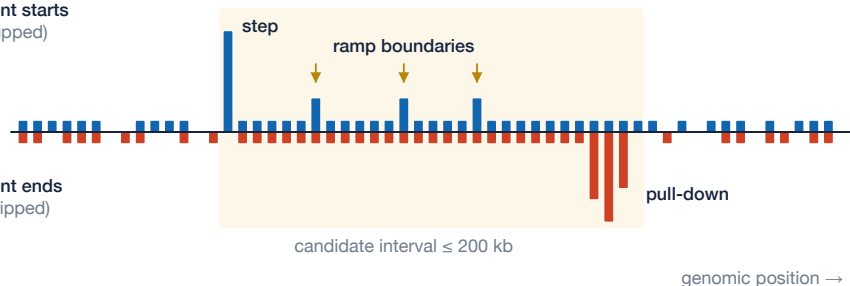


a Two-state detection of copy-number-altered intervals

alignment starts
(front-clipped)

alignment ends
(back-clipped)



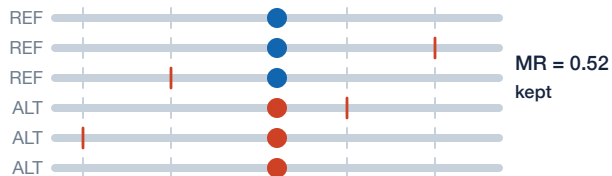
state	rule
open, step	≥ 5 front-clips at one position
open, ramp	front-clips $>$ back-clips, below that
extend	excess $c > 30$, 30 kb look-ahead
close, step	back-clips $\geq n/2$ ($n \geq 10$), else 5
close, ramp	back-clips $\geq c/4$

b Mismatch load separates true heterozygotes from paralogues

R_m = alternate alleles carried by one read inside the interval

$$MR = \frac{\text{mean } R_m (\text{ALT})}{\text{mean } R_m (\text{REF}) + \text{mean } R_m (\text{ALT})} \geq 0.7 \rightarrow \text{remove the site from every read}$$

true heterozygous site



parologue-specific difference



grey ticks, reference allele

red ticks, alternate allele at another heterozygous site